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Technical Design and Data Model

Mizzey Operations Platform. The stack and the reasoning behind it, what WooCommerce carries and what has to be written, the module inventory, the data model, the integrations and the operating cost.

The Stage 1 technical design, satisfying PRE-01 and PRE-02. Scope is defined by the Feature Register MS-ANX-2026-001 and by nothing else, including this document.



CLIENT

Mizzey.com

VERSION

1.0

DATE

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STATUS

■ For client approval

PREPARED BY

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COMPANION DOCUMENT

Feature Register & Scope Definition, MS-ANX-2026-001

1. What this document is

Two of the pre development deliverables in Annex A Part Seven are answered here. **PRE-01**, the technical design and architecture document, and **PRE-02**, the technology stack proposal with its reasoning and its expected operating cost.

It also answers a question the Feature Register raised and did not settle in engineering terms: your requirements document section 19.1 required the developer to propose a stack and justify it. Annex A section 2.3 states the choice. This document states what that choice actually gives you, what it does not, and what has to be built because of the difference.

It changes no scope. Where this document and Annex A differ, Annex A governs.

2. The layers

`` WordPress the content platform, users, roles, media, i18n WooCommerce the commerce engine: products, stock, cart, orders, coupons, refunds CoreX the framework layer: container, config, routing, middleware, admin shell, forms, blocks, mail, media, CLI Mizzey the store: theme, commerce rules, Operations Console, storefront controls, integrations, migration tooling ``

Each layer may depend on the layer below it and never on the layer above. The Mizzey layer is the only one written for this client. **CoreX is pre existing and maintained independently of this project**, under its own licence, as Annex A section 2.4 records.

3. Why WordPress and WooCommerce

Developer technical decision, in answer to your requirements document section 19.1. Your document names no technology, and this choice is the developer's rather than yours.

The reasoning in Annex A section 2.3 stands. Restated in engineering terms:

The expensive, dangerous parts are already written and already proven. Order state machines, stock decrement under concurrency, tax and discount arithmetic, partial refunds against a gateway, coupon eligibility. These are where bespoke commerce projects lose their budget, and where a defect costs real money rather than embarrassment. Section 6 sets out exactly how much of the register they cover.

Both recommended providers publish maintained integrations for it. Paymob and Bosta each ship a WooCommerce plugin. Integrating against raw APIs instead would add cost and risk for no gain (PAY-13, SHIP-08).

It does not lock you in. The data model is standard and exportable, hosting is yours, and the code is delivered to a repository you own (HND-01, MIG-25).

4. Why CoreX, and what state it is actually in

Developer technical decision. Not referred to in your requirements document.

CoreX is the developer's pre existing framework layer. **This section states its real condition rather than its intended one**, because the honest version is the one that can be planned against.

COREX CAPABILITY	STATE	WHAT MIZZEY USES IT FOR
Container, layered configuration, routing, middleware, security, cache, jobs, notifications	Stable	The spine of every Mizzey module
Admin product: settings, data models, submissions, access, operations, insights	Stable	The Operations Console and the storefront controls are built on this
Forms: schema, validation, flow builder, submission pipeline, file uploads	Stable	Return requests, contact, import wizard
Server rendered blocks	Stable	Storefront and campaign page composition
Brand tokens and visual foundation	Stable	The design system the interface design is implemented into
Header, mobile navigation, mega menu, footer system	Stable	NAV-01 to NAV-10, SSC-07, SSC-08
Admin experience: shell, data explorer, settings	Stable	Every custom admin screen
Mail: templates, routes, queue, attempt log	Stable	Every notification in Epic E12
Media: WebP pipeline behind an activation gate	Stable	NFR-02 image optimisation
Client site delivery: make:site , sites/ layout, dist builder, deployment	Stable	How Mizzey is generated, built and deployed
CLI generators	Stable	Module scaffolding
WooCommerce kit	Not built	Nothing. See below

NOTE**The one that matters: there is no CoreX commerce kit**

corex-kit-woo is four files, being a blueprint, a gate and a service provider, and its own documentation describes it as reserving the seam rather than being a store kit. It is milestone M9 on the CoreX roadmap and it is not started.

The practical consequence for this project. Every storefront template, every commerce screen, every commerce rule in this build is Mizzey specific work. CoreX contributes the framework beneath them, not the store itself. Nothing in Annex A promised otherwise: section 2.4 lists what CoreX provides as architectural structure, custom application modules, the Operations Console, the storefront management controls, the integration adapters and the migration tooling, and every one of those is described there as something built for Mizzey on CoreX.

This is recorded here so that the estimate can be read honestly. The Delivery Backlog MS-BLG-2026-020 prices the custom epics at up to 1.7 times the configuration epics for exactly this reason.

What CoreX does remove is the six to eight weeks a project of this shape normally spends building plumbing: dependency injection, environment layered configuration, an admin shell that looks like one product, a forms pipeline, a mail queue with an attempt log, an audit trail, a WebP pipeline, a release and deployment path. Those are present, tested and in use.

PART TWO: WHAT IS BOUGHT AND WHAT IS BUILT

5. How to read the next two sections

Annex A contains 598 contracted requirements. The single most useful thing this document can do is say, for each area, whether the behaviour arrives with WooCommerce or has to be written.

Four classifications are used, and they are the same ones the Delivery Backlog estimates against.

CLASSIFICATION	MEANING
Native	WooCommerce implements the behaviour. The work is configuration, the theme layer, and proving it
Extend	Woo provides the mechanism and a hook. The required behaviour is a filter or an action on top
Plugin	A maintained third party component does it. Named, with its licence cost where it has one
Build	Nothing provides it. A Mizzey module, written on CoreX

6. Where WooCommerce carries the contract

These are the areas where the platform choice pays for itself. **They are also the areas with the highest cost of failure**, which is the point.

AREA	REQUIREMENTS	HOW	NOTE
Products, variants, SKU, barcode, stock, pricing	ADM-20 to ADM-45, ENT-01, ENT-02	Native	Variable products cover every variant requirement
Categories, brands, attributes, collections	ADM-55 to ADM-57, ENT-03	Native	Brands and collections as taxonomies
Cart and checkout mechanics	CART-01 to CART-05, CHK-01 to CHK-12	Native	Address fields need Egyptian structure, section 7
Order lifecycle and statuses	ORD-01 to ORD-12	Native	Two custom statuses added, section 7
Price snapshot on the order	BR-005, ENT-08, AC-09	Native	Order line items store the price charged. This was the highest risk business rule in the register and the platform already guarantees it

AREA	REQUIREMENTS	HOW	NOTE
Payments, tokenised hosted flow, no card storage	PAY-02, PAY-03, PAY-14, NFR-06	Native plus provider plugin	Paymob is PCI DSS Level 1 and hosted
Full and partial refunds against the gateway	PAY-07, PAY-08, PAY-17, ADM-91	Native	Woo refund API enforces the paid amount ceiling, which is BR-006
Stock validation and decrement under concurrency	BR-003, CHK-12, AC-05	Native	Reduce stock at payment, restore on cancel, which is BR-007
Coupons: percentage, fixed, product and category limits, usage limits, expiry	PROMO-01 to PROMO-15, PROMO-18, PROMO-19	Native	Stacking control is not native, section 7
Guest checkout and account creation	AUTH-07, FIX-09, BR-008	Native	Linking a guest order to a later account is Extend
Customer accounts, addresses, order history	ACCT-01 to ACCT-08, ENT-05	Native	My Account endpoints
Reviews with moderation	REV-01 to REV-04, ENT-14	Native	WordPress comment moderation
Shipping zones and flat rates	SHIP-01, SHIP-15	Native	Zones by governorate as shipping zones
Tax classes	ADM-36	Native	Configured once OD-09 is answered
REST API and webhooks	INT-16	Native	The adapter layer wraps them

7. Where WooCommerce stops, and what gets written

This is the real content of the build. Each row is a place where the register requires behaviour the platform does not provide. None of them is a blocker. All of them are known quantities, and each is named so that no one discovers it in week nine.

#	REQUIREMENT	WHAT WOODOES	WHAT GETS BUILT	CLASS	SIZE
1	Guest cart survives sign in and merges without duplicates (CART-13, CART-14, CART-15, CART-18, AC-18)	On login, Woo replaces the session cart with the persistent cart, or keeps the session cart and discards the saved one, depending on path. It does not merge	A merge on woocommerce_cart_loaded_from_session : union by product and variation, quantities combined, capped at available stock, prices re validated, and the customer told what changed	Build	Medium, and the most delicate logic in the storefront
2	Free shipping on two or more items (BR-002, FIX-07, AC-01, AC-02)	Free shipping supports a minimum order amount or a coupon. It has no item count condition	A shipping method condition on woocommerce_package_rates , reading a configured rule, plus the cart progress prompt (CART-07)	Extend	Small
3	Promotion stacking and priority (BR-004, PROMO-16, PROMO-17, ADM-111)	Coupons combine freely. There is no priority and no stacking policy	A promotion resolver that evaluates eligible promotions, applies the configured priority, enforces the stacking policy, and itemises what was applied	Build	Medium
4	Welcome discount on registration (BR-001, AUTH-06, FIX-06, AC-03)	Nothing automatic	Rule evaluation at registration and at checkout, with value, cap, expiry and eligibility all configured	Build	Small

#	REQUIREMENT	WHAT WOOL DOES	WHAT GETS BUILT	CLASS	SIZE
5	Returns and refunds workflow (RET-01 to RET-10, ADM-120 to ADM-125, E13, E20)	Wool has refunds. It has no returns workflow at all : no request, no review, no decision, no receipt, no condition, no status the customer can see	A full RMA module: request capture with evidence, eligibility against the configured window, a review queue, approve, reject or partially approve with a recorded reason, receipt and condition, refund against the original payment, and customer visible status at every stage	Build	Large
6	Catalogue migration tooling (MIG-01 to MIG-25, Section U)	Wool ships a basic CSV importer with fixed column handling, no dry run, no saved mappings, and no safe re run semantics	A mapping driven import engine: encoding detection verified against Arabic, a column mapping interface with saved profiles, preview, a dry run that writes nothing, safe re runs that update by SKU, parent and child rows converted into variable products, background processing, a row level error report, and an import history	Build	Largest single item in the project
7	The Operations Console (ADM-140 to ADM-161, Section G10)	Wool has separate list screens and a basic status report	A single work queue page: orders requiring action, stock requiring attention, today figures, saved views, quick actions, an inline price and stock grid, bulk operations, and search by customer phone	Build	Large
8	Self service storefront control (SSC-01 to SSC-29, Section G11)	Wool controls products. It does not control the storefront around them	Content control for home sections, category and brand pages, collections, navigation, footer, campaign pages, all per language, with preview before publish and revert	Build	Large
9	Arabic search that finds what was typed (SRCH-06, SRCH-07)	WordPress search is a LIKE query. It has no synonyms, no correction and no relevance	A curated synonym and correction list, admin managed, applied at query time across the main categories and brands including Franco Arabic spellings	Build	Medium. Quality is bounded by the list, which is why the register records it P1-L
10	Returned to origin is never Delivered (ORD-13, SHIP-17, AC-16)	Wool has no such status, and the carrier integration has reported cases of mapping a returned parcel to delivered	Two registered order statuses, plus an explicit carrier status map that is documented and visible in admin, and an unrecognised carrier status that changes nothing and raises for a human	Extend	Small, and it prevents an expensive error

#	REQUIREMENT	WHAT WOOL DOES	WHAT GETS BUILT	CLASS	SIZE
11	Cash on delivery reconciliation (SHIP-16, ORD-14)	Wool marks a COD order paid on completion. Collected and remitted are the same thing to it	Collected and remitted tracked separately per order, with the difference reportable	Build	Small
12	Duplicate order and duplicate charge protection (PAY-05, NFR-07, AC-06, AC-07)	Some protection exists, and gateway callback idempotency is the integrator responsibility	An idempotency key on order placement, and a processed callback record so a repeated provider callback is a no operation	Extend	Small, and non negotiable
13	Wishlist (WISH-01 to WISH-04, WISH-07, ENT-07)	Nothing	A wishlist for accounts and a session wishlist for guests that transfers on sign in, plus recently viewed	Build	Small
14	Audit log (ADM-132 to ADM-134, ROLE-10, ENT-17)	Nothing	Actor, action, entity, before and after values, timestamp, searchable, and not deletable from the interface	Build on the CoreX audit seam	Medium
15	Staff roles beyond the Wool defaults (ROLE-03 to ROLE-09, ADM-130)	Wool ships Shop Manager and Customer only	Six role definitions with granular capabilities, and the negative tests that prove a role cannot reach what it must not	Build	Medium
16	Google sign in (AUTH-10, AUTH-13, AC-19)	Nothing	OAuth sign in, and identity resolution so one email is always one account	Plugin or Build	Small
17	Multilingual product and content (FIX-04, NFR-04, SSC-12, SSC-21)	Nothing. WordPress is single language	A translation plugin, plus the right to left layout system and locale formatting	Plugin , and it has a licence cost. Section 11	Medium, and it touches everything
18	Product feeds and campaign attribution (MKT-05, MKT-06, MKT-09)	Nothing	Meta and Google Merchant feeds, and campaign parameters preserved to the order record	Plugin plus Extend	Small
19	Redirect manager and missing page log (MKT-13, MKT-14)	Nothing	Admin managed permanent redirects with loop detection	Plugin	Small

#	REQUIREMENT	WHAT WOOCOMMERCE DOES	WHAT GETS BUILT	CLASS	SIZE
20	One step account creation after ordering (AUTH-11)	Woo offers account creation at checkout, not after it from the confirmation with no re entry	Password only account creation from the order confirmation, with the order linked securely	Build	Small

NOTE

The answer to the question this section exists for. Yes, the specification is implementable on WooCommerce, and the platform carries the requirements whose failure would cost money.

Roughly a third of the effort is in five Build areas that Woo contributes nothing to, being returns, the Operations Console, the storefront controls, the migration tooling and the search layer. That proportion is normal for a store with an operational contract of this depth, and it is what the estimate is built on.

PART THREE: THE BUILD

8. Module inventory

Every module is a CoreX service provider inside the Mizzey site plugin, generated with **wp corex make:site Mizzey**. Each owns its data, exposes its own interfaces, and depends on the platform rather than on its siblings.

MODULE	OWNS	EPICS
Mizzey\Storefront	Templates, blocks, navigation, home sections, campaign pages	E01, E02, E03, E04, E22
Mizzey\Cart	Cart continuity, merge, free shipping progress, cross sell	E05
Mizzey\Identity	Google sign in, account resolution, one step creation, session policy	E06
Mizzey\Checkout	Egyptian address structure, order placement, idempotency	E07
Mizzey\Payments	The payment abstraction and the Paymob adapter	E08
Mizzey\Shipping	The carrier abstraction, the Bosta adapter, status mapping, COD reconciliation	E09
Mizzey\Promotions	The promotion resolver, stacking, priority, welcome discount, business rules	E14
Mizzey>Returns	The RMA workflow, customer side and staff side	E13, E20
Mizzey\Operations	The Operations Console, work queues, bulk actions, saved views	E15, E18
Mizzey\Catalogue	Product controls, per language content, cost, inventory	E16, E17
Mizzey\Migration	The mapping driven import and export engine	E23
Mizzey\Access	Roles, capabilities, audit log	E19, E21
Mizzey\Insight	Dashboard, reports, analytics events, feeds	E24, E25, E26
Mizzey\Notify	Notification routing on the CoreX mail queue	E12

9. Data model

Annex A Section K names nineteen entities. Fourteen exist in WooCommerce already. **Five are new tables**, and one is a new field on an existing entity.

ENTITY	WHERE IT LIVES	NOTE
ENT-01 Product, ENT-02 Product Variant	Woo products and variations	Native
ENT-03 Category, Brand, Collection	Woo taxonomies	Brand and collection as custom taxonomies
ENT-05 Customer, Address	WordPress users plus Woo customer data	Egyptian address fields added
ENT-06 Cart, Cart Item	Woo session and persistent cart	The merge logic is Mizzey, the storage is Woo
ENT-08 Order, Order Item	Woo orders, HPOS tables	Price snapshot native. Attribution fields added for MKT-09
ENT-09 Payment	Woo order meta plus a payment record	Transaction reference, status, amount, timestamps
ENT-10 Shipment	New table	Carrier, tracking, rate, status, and the collected versus remitted pair for SHIP-16
ENT-11 Promotion, Coupon	Woo coupons plus a promotion rules table	Stacking, priority and conditions the coupon schema cannot hold
ENT-12 Return Request, Return Item	New table	Reason, status, attachments, decision, actor, timestamps
ENT-13 Refund	Woo refunds, linked to the return	Native amount ceiling
ENT-14 Review	WordPress comments	Native moderation
ENT-15 Notification	CoreX mail attempt log	Channel, template, status, provider reference
ENT-16 Admin User, Role, Permission	WordPress roles and capabilities	Six role definitions
ENT-17 Audit Log	New table	Append only from the interface, per ADM-133
ENT-19 Import Run	New table	File, mapping, row counts, outcome, actor, timestamp
ENT-07 Wishlist	New table	Account and guest
ENT-04 Inventory Transaction	Second release	Stock levels are native and live at launch
ADM-27 Product cost	New field on product and variation	Captured from day one so margin reporting can be built later on real history

Two design rules apply across all of it.

Nothing that WooCommerce owns is duplicated. Where Woo holds the truth, Mizzey reads it rather than keeping a copy that can drift.

Every new table carries actor and timestamp. The audit requirement is not a feature bolted on at the end; it is a column pair on everything that changes.

10. Integrations

Annex A INT-16 requires every external integration behind an adapter with retries, logging and webhook handling. That is a single CoreX pattern applied five times.

INTEGRATION	ADAPTER	FAILURE BEHAVIOUR
Paymob	Payments\Gateway interface, Paymob adapter over its Woo plugin	Callback signature verified before anything is acted on. A failed verification is rejected, logged and raised. A repeated callback is a no operation
Bosta	Shipping\Carrier interface, Bosta adapter	A dispatch failure is reported per order and is safe to retry for the failures only. An unrecognised carrier status changes nothing
Transactional email	CoreX mail with a queue and an attempt log	Fail closed by default. A send that cannot be made is visible, not silent
Analytics and Tag Manager	A data layer emitter, tags deployed through Tag Manager only	Consent aware. No third party edits site code, per MKT-23
Product feeds	Scheduled generation	A feed that fails to validate is reported rather than published

The abstraction is why a provider change is not a rebuild. Annex A section 2.1 commits to accommodating a different payment provider without a change fee where the integration requirements are materially equivalent, and this is the mechanism that makes that commitment safe to give.

11. Third party components and their annual cost

PRE-02 requires the expected operating cost, and this is the part of it the developer controls. Hosting and provider fees are separate and are yours (Annex A CR-14, R-14).

COMPONENT	PURPOSE	LICENCE	APPROXIMATE ANNUAL COST
Translation plugin	Multilingual products, categories and content, per FIX-04 and SSC-21	Commercial	Roughly 100 USD or EUR per year
Google sign in	AUTH-10	Free tier available	Nil, or low
Redirect manager	MKT-13	Free	Nil
Two factor authentication for admin	ADM-131	Free	Nil
Product feeds	MKT-05, MKT-06	Free options available	Nil
WordPress, WooCommerce, CoreX	The platform	Open source, and the Corex licence	Nil

NOTE**The multilingual licence, stated plainly**

A fully multilingual WooCommerce store requires a commercial translation plugin. The free options do not translate product data reliably, and FIX-04 places a fully supported Arabic storefront in the first release as a fixed decision, not an option.

This is not new scope and it is not a surprise charge from the developer: Annex A R-14 already records that certain commercial components require annual licences held in your name, and that these are your ongoing operating costs. **It is named here with a figure so that it reaches your budget now rather than at launch**, and so that the licence is bought in your name and stays yours.

It is the only component in this build with a meaningful recurring licence cost.

12. Environments, deployment and recovery

ENVIRONMENT	PURPOSE	PER ANNEX A
Local	Development on WAMP	
Staging	Every stage gate is reviewed here. Risky changes are never tested on the live site	NFR-09
Production	Provisioned once the hosting instruction and any required authorisation are in place	CR-10, R-03

Deployment is the CoreX dist builder and verifier, producing a shared host deployable artefact, driven from the repository. **Rollback is documented and tested before go live**, not written at go live (HND-08).

Backups cover the database and media, to the agreed retention, and **the restore procedure is performed at least once rather than only written** (NFR-08, HND-05). A restore procedure that has never been run is a document, not a recovery plan.

13. Security posture

The controls in NFR-05, NFR-06, AUTH-15, AUTH-16 and ADM-131 are implemented as follows, and none of them is optional.

CONTROL	HOW
Card data	Never touches the server. Provider hosted, tokenised (PAY-02, PAY-03)
Payment callbacks	Cryptographically verified before being acted on (PAY-16)
Passwords	WordPress secure storage, never recoverable in readable form
Sessions and login	Rate limited, documented expiry, automated account creation protected against
Password reset	Single use and time limited (AUTH-16)
Admin	Two factor available and requirable per role (ADM-131)
Authorisation	Capability checks on every admin action, with negative tests, per AC-11
Transport	HTTPS throughout
Input	Validated and escaped at the boundary
Secrets	Environment configuration, never in the repository
Dependencies	Version pinned, updated on a schedule, regression tested on staging before production (R-08)
Audit	Sensitive operations logged and not deletable from the interface (ADM-133)

No guarantee of immunity from all future vulnerabilities is given, and none can be. The Services Agreement says the same.

14. Performance

MEASURE	APPROACH
Images	WebP through the CoreX pipeline, served only when the conversion passes its gate, lazy loaded
Caching	Page and object caching, with the cart, checkout and account correctly excluded
Delivery	Content delivery network in front of static assets
Queries	Product listing and search queries reviewed against the real catalogue size, not against test data
Measurement	Measured on staging before launch and recorded

Sizing follows the stated assumption at OD-15: **up to 5,000 products and 300 orders per day**, with headroom for short campaign peaks. Growth beyond that is absorbed by resizing infrastructure rather than re implementing the core (NFR-13), bounded by FIX-10.

No performance, conversion, traffic or ranking outcome is warranted (EX-31).

PART FOUR: RISKS THIS DESIGN CARRIES

15. Technical risks arising from these choices

Annex A Part Four holds the commercial and compliance risks. These are the engineering ones, and they are the developer's to manage unless the owner column says otherwise.

ID	RISK	OWNER	MITIGATION
TR-01	The cart merge is subtle and its failures are silent. A customer who loses a cart does not report a defect, they leave	Developer	Automated tests over every path in AC-18, and the merge treated as a key requirement rather than a convenience
TR-02	The carrier integration has a known status mapping defect, with reported cases of a returned parcel showing as delivered	Developer	An explicit documented map, an unrecognised status that changes nothing, and AC-16 tested against real carrier responses in Stage 4 rather than against documentation
TR-03	The migration tool is the largest single build and its inputs are outside the developer control. Amazon report formats change, and the complete listing report is not enabled by default	Shared	The real sample file is required before the build begins (CR-06). The engine is mapping driven rather than format specific, so a changed format is a new mapping and not a new tool
TR-04	The translation plugin becomes a dependency of everything a customer reads. A major version change can affect the whole storefront	Developer	Version pinned, upgraded only on staging with a full bilingual regression pass
TR-05	Multilingual content quality is bounded by the client content, not by the system	Client	CR-15. The mechanism is delivered whether or not the words arrive, and a missing translation is reportable before launch rather than discovered by a customer
TR-06	Concurrency on the last unit of stock	Developer	Woo handles the decrement; the tests prove it under concurrent purchase, per US-27-04
TR-07	Third party plugin surface increases the update and security burden after warranty	Client after handover	The plugin count is kept deliberately low, section 11. R-12 applies: an unmaintained site becomes progressively more exposed
TR-08	CoreX has no commerce kit, so there is no reference implementation to follow for the storefront and commerce screens	Developer	Accepted and priced. The Delivery Backlog carries the custom leverage factor for exactly this

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